

L9b: Pricing American Style Options Contracts using the Cox, Ross, Rubinstein (CRR) Binomial Model

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced European options, which can be exercised only at expiration. However, standard options on stocks and ETFs are American-style and can be exercised at any time. What is the right to exercise early worth?

Today's concepts

- Distinguish between European and American options: exercise only at expiration, or at any time before or on expiration.
- Apply the binomial model to price American options: build CRR trees and compute premiums with the decision rule for early exercise.
- Estimate the premium of American options by simulation: Longstaff–Schwartz Monte Carlo, regressing the value of holding on the share price.

Lectures and Examples

Lecture:
CHEME-5660-L9b-Lecture-American-Derivatives-Intrinsic-Extrinsic-Value-Fall-2026.ipynb

Example 1:
CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Example 2:
CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Review on your own:
CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Company Profile: Options Clearing Corporation

OCC stands between the buyer and the seller of every U.S. listed option. It opened in 1973 as CBOE Clearing Corporation and became OCC in 1975.

- Exercise and assignment: OCC assigns each early exercise at random to a clearing member with a short position in the same contract.
- Automatic exercise at expiration: since 1977, an expiring equity option at least 0.01 USD/share in the money is exercised unless the holder instructs otherwise.
- Monte Carlo margin and binomial trees: since 2006, STANS simulates 10,000 two-day scenarios and reprices American options on a binomial tree.

Connection to today's lecture: a seller of an American put can be assigned once exercising now beats holding, so a European-premium margin would leave out the early-exercise premium.

American Style Options Contracts

American-style contracts can be exercised at any time before or on the expiration date T. Previously, we wrote the American premium as:

$$\mathcal{P}_\star^{\text{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V_\star(K, S(\tau))\Big) \geq \mathcal{P}_\star^{\text{EU}} \qquad \text{for } \star \in \{c, p\}$$

- Stopping time: τ uses only the information available at that time.
- Discounting: $\mathcal{D}_{\tau,0}^{-1}(g_y)$ brings the payoff at strike price K back to today.
- At least the European premium: the holder can always wait.

The standard pricing method is the Cox, Ross, Rubinstein (CRR) binomial model.

Reference: Cox, Ross, and Rubinstein (1979). Option pricing: A simplified approach. J. Financial Economics 7(3): 229.

CRR Binomial Lattice Model

The share price moves up by u with risk-neutral probability q, or down by d:

$$
\begin{aligned}
\mathcal{D}_{1,0}(g_y) \cdot S_0 &= \mathbb{E}_{\mathbb{Q}}\left(S_1\right) && \text{grow at the risk-free rate } g_y \text{ under } \mathbb{Q} \\
&= q \cdot (u \cdot S_0) + (1-q) \cdot (d \cdot S_0) && \text{up with probability } q\text{, down with } (1-q) \\
&= S_0(qu + (1-q)d) && \text{factor out } S_0 \\
&= S_0(qu + d - qd) && \text{expand } (1-q)d \\
&= S_0(d + q(u-d)) && \text{collect the } q \text{ terms}
\end{aligned}
$$

$$
\implies \boxed{q = \frac{\mathcal{D}_{1,0}(g_y) - d}{u - d}} \qquad \text{divide by } S_0 \text{ and solve for } q
$$

where S_0 is today's share price and $R_f = \mathcal{D}_{1,0}(g_y) = e^{g_y \Delta t}$ for one step Δt.

CRR Up and Down Factors

With $ud = 1$, let $a = \ln u = -\ln d$. Match the growth rate $g = \pm a/\Delta t$ to GBM:

$$\frac{(2p-1)\,a}{\Delta t} = \mu_g \qquad \text{match the mean growth rate}$$

$$\frac{4p(1-p)\,a^2}{\Delta t^2} = \frac{\sigma^2}{\Delta t} \qquad \text{match the variance}$$

$$\frac{a^2}{\Delta t^2} - \mu_g^2 = \frac{\sigma^2}{\Delta t} \qquad \text{use } 4p(1-p) = 1-(2p-1)^2 \text{ and the mean}$$

$$a^2 = \sigma^2 \Delta t + \mu_g^2 \Delta t^2 \qquad \text{solve for } a^2$$

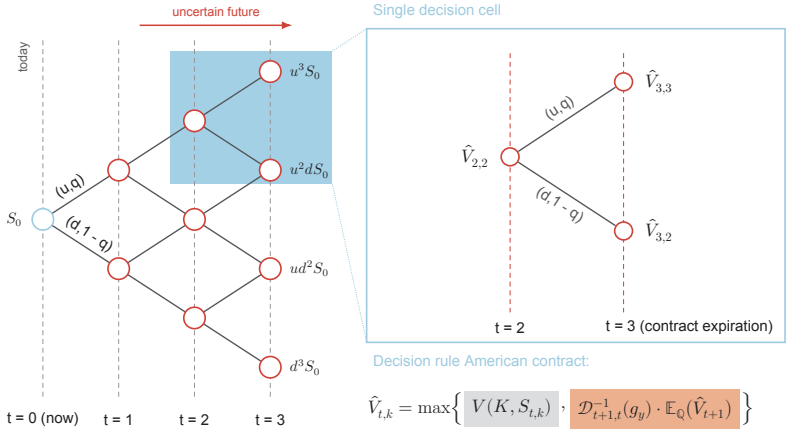
$$\implies \boxed{u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}} \qquad \text{keep the first term, so } a = \sigma\sqrt{\Delta t}$$

where p is the real-world up probability, μ_g the mean growth rate, and σ the volatility.

Where do u and d come from? advanced/crr_factors/CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Decision-Rule for American Options

At each node, the American holder decides to exercise now or hold for the future.



The Early-Exercise Premium

Working backward from expiration, the American holder takes the larger of exercising and holding, while the European holder can only hold:

$$\hat{V}^{\mathsf{AM}}_{t,k} = \max\Big\{ \underbrace{V(K, S_{t,k})}_{\text{exercise now}},\ \underbrace{\mathcal{D}^{-1}_{t+1,t}(g_y) \cdot \overbrace{\left(q\,\hat{V}^{\mathsf{AM}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{AM}}_{t+1,k}\right)}^{\mathbb{E}_{\mathbb{Q}}}}_{\text{hold for the future}} \Big\}$$

$$\hat{V}^{\mathsf{EU}}_{t,k} = \mathcal{D}^{-1}_{t+1,t}(g_y) \cdot \overbrace{\left(q\,\hat{V}^{\mathsf{EU}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{EU}}_{t+1,k}\right)}^{\mathbb{E}_{\mathbb{Q}}}$$

where $\hat{V}_{t,k}$ is the contract's value with k up moves in t steps, $S_{t,k} = S_0 u^k d^{t-k}$, and $\mathcal{D}^{-1}_{t+1,t}(g_y) = e^{-g_y \Delta t}$. The premium is $\mathcal{P} = \hat{V}_{0,0}$.

Early-exercise premium: $\mathcal{P}^{\mathsf{AM}} - \mathcal{P}^{\mathsf{EU}} \geq 0$.

Pseudocode: Pricing on the CRR Lattice

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, and the number of time steps h. Compute $\Delta t = T/h$, u and d, q, and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

1. Build the share price lattice: $S_{t,k} \leftarrow S_0 u^k d^{t-k}$ for each level $t = 0, 1, \ldots, h$ and node $k = 0, 1, \ldots, t$.
2. At expiration, set each node value to the payoff: $\hat{V}_{h,k} \leftarrow V(K, S_{h,k})$ for $k = 0, 1, \ldots, h$.
3. For each level $t = h-1, h-2, \ldots, 0$ **do**:
   - For each node $k = 0, 1, \ldots, t$ **do**:
     a. Compute the value of holding for the future: $H_{t,k} \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot (q\,\hat{V}_{t+1,k+1} + (1-q)\,\hat{V}_{t+1,k})$.
     b. If the contract is American, **then**: $\hat{V}_{t,k} \leftarrow \max\{V(K, S_{t,k}),\ H_{t,k}\}$. Otherwise, $\hat{V}_{t,k} \leftarrow H_{t,k}$.
4. The premium is the value at the root: $\mathcal{P} \leftarrow \hat{V}_{0,0}$.

A Hypothetical Example from Hull

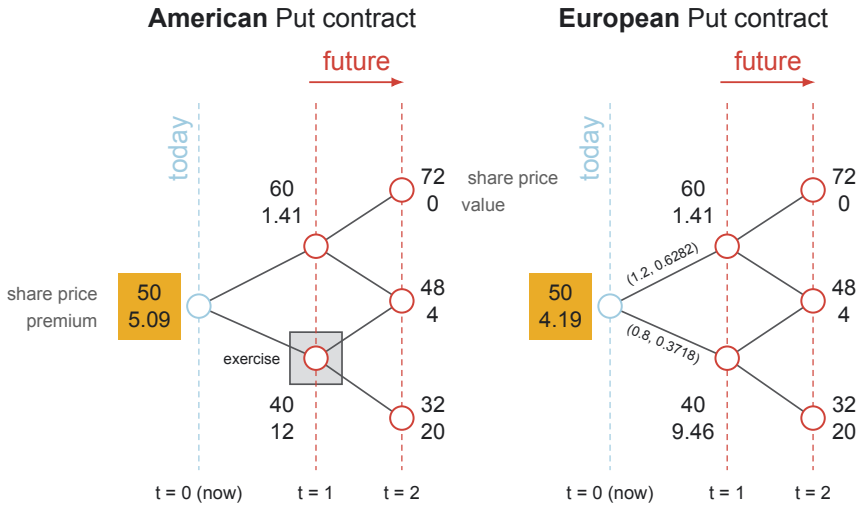
Consider a put contract.

- Contract: the strike price is $K = 52$ USD/share on a stock whose current price is $S_0 = 50$ USD/share.
- Lattice: two time steps of 1 year each. In each period, the stock price moves up by a factor of $u = 1.2$ or down by a factor of $d = 0.8$.
- Pricing inputs: the continuously compounded risk-free rate is $g_y = 5\%$, so the risk-neutral probability of an up move is $q = (e^{0.05} - 0.8)/(1.2 - 0.8) = 0.6282$.

What is the price of the American and European put option contracts?

Source: Chapter 13 of Hull, Options, Futures, and Other Derivatives, 9th edition (2015).

American versus European Put on One Lattice



Early Exercise in the Hull Example

At the down node at $t = 1$, where $S_{1,0} = 40$ (all values in USD/share):

$$\underbrace{V(K, S_{1,0}) = 52 - 40 = 12}_{\text{exercise now}} \; > \; \underbrace{\mathcal{D}_{2,1}^{-1}(g_y)(q \cdot 4 + (1-q) \cdot 20) = 9.46}_{\text{hold for the future}}$$

so the American holder exercises. That one decision raises the premium:

$$\boxed{\mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}} = 5.09 - 4.19 \approx 0.90}$$

Let's compute every node of this lattice, then apply the same method to contracts from the course's options archive.

Review on your own: CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

We'll work through in lecture: CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Monte Carlo Pricing of American Options

We simulate share price paths under $\mathbb{Q}$, then estimate the value of holding at each date by regressing the discounted cash flows from holding on the current share price across paths. Like the lattice, Longstaff and Schwartz work backward from expiration:

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, the number of exercise dates h after today, and the number of paths M. Compute $\Delta t = T/h$ and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$ (expiration), where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths, those with $V(K, S_i(t)) > 0$, fit the value of holding $\hat{C}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{C}_t(S_i(t))$, **then**: exercise, so $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let's price an American put by simulation and compare it with the lattice and the European premium. We apply two rules to the same simulated paths:

Rule at each date	Estimates
Longstaff–Schwartz: exercise when $V(K, S_i(t)) > \widehat{C}_t(S_i(t))$	$\mathcal{P}^{\text{AM}}$
Hold on every path	$\mathcal{P}^{\text{EU}}$

$$\boxed{\text{early-exercise premium} \approx \mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}}}$$

Example 2: CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Optional Advanced Material

CRR factors: advanced/crr_factors/
CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

In this notebook, we derive the up and down factors of the CRR model by matching the mean and variance of the lattice's one-step growth rate to those of geometric Brownian motion. We then show that the lattice still matches geometric Brownian motion when we price with the risk-neutral probability, and find how small the time step must be to keep that probability between zero and one.

Notebooks: lectures/week-9/L9b/advanced, also linked from the lecture notebook.

Summary

Today we priced American options with the CRR binomial model and Monte Carlo simulation.

Key takeaways

- Binomial trees price American options: at each node, the holder exercises or holds, so backward induction computes the premium.
- The decision rule sets the early-exercise premium: in Hull's put, exercising at the one node where it beats holding accounts for the whole gap between the American and European premiums.
- Simulation estimates the same premium: Longstaff–Schwartz works backward along simulated paths, regressing the value of holding on the share price. Holding on every path gives the European premium.

The early-exercise premium is what the right to exercise before expiration is worth, and the lattice and simulation give two ways to compute it.